The Future of IoT in Smart Cities

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Abstract

More than half of the world's people now live in cities, and those cities have to keep water, power, roads and waste collection running while the population keeps growing. The Internet of Things (IoT) is one of the main tools they are using to cope. This case study looks at where smart-city IoT is today and where it is going. It sets out a simple layered architecture, goes through three real examples (Barcelona, Singapore and Songdo), and then looks at the technologies likely to matter next: edge computing, 5G and 6G, and digital twins. It also covers the problems that can sink a project, mainly security, privacy, cost and public trust. The main point I draw from the evidence is that adding more sensors matters less than sharing the data between services, protecting it, and starting from what residents actually need.

1. Introduction and market context

A smart city uses connected devices and data analysis to run urban services more efficiently. A street lamp that dims when the road is empty, a bin that reports when it is full and a valve that closes when rain is forecast are all the same idea. A device senses something, a network carries the reading, a platform analyses it and something happens as a result. Early projects were mostly isolated pilots. Newer ones try to join many services together so that the city can be managed as one system.

The money going into this area is large. One market report puts IoT in smart cities at US$345.5 billion in 2024 and forecasts US$987.9 billion by 2030, which is about 19.1% growth a year [1]. A wider forecast for the whole smart-cities market reaches US$3,728.3 billion by 2030, with Asia Pacific growing fastest at 29.2% a year [2]. Traffic management and smart energy grids are named as the biggest segments, helped by 5G, edge computing and AI [1]. Different analysts draw the market boundary in different places, so these numbers show direction and rough scale rather than exact values. Figure 1 shows the forecast and the reported growth rates.

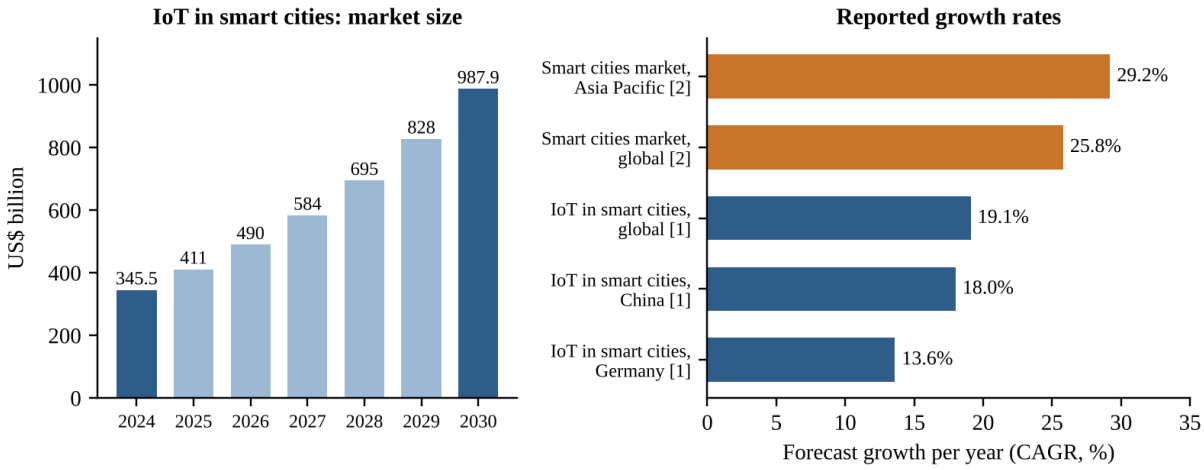


Figure 1. Left: IoT-in-smart-cities market size, with 2024 and 2030 taken from the report [1] and the years between interpolated at its 19.1% growth rate. Right: growth rates from two separate market reports [1][2], which use different definitions and periods.

2. Architecture of a smart-city IoT system

Most deployments can be described with four layers and a security plane that runs through all of them (Figure 2). At the bottom are the devices: environmental and traffic sensors, smart meters, cameras and actuators such as valves and signal controllers. Above them is the network, and the choice of network depends on the job. Battery-powered meters and bin sensors suit low-power networks like NB-IoT and LoRaWAN, video needs Wi-Fi or fibre, and moving or time-critical devices suit 5G. The platform layer stores and analyses the data and shares it through open interfaces. The application layer turns the results into services for city staff and residents. Edge computing sits across these layers. Doing some of the processing near the devices cuts delay and bandwidth use, and it keeps raw data local, which helps privacy [11].

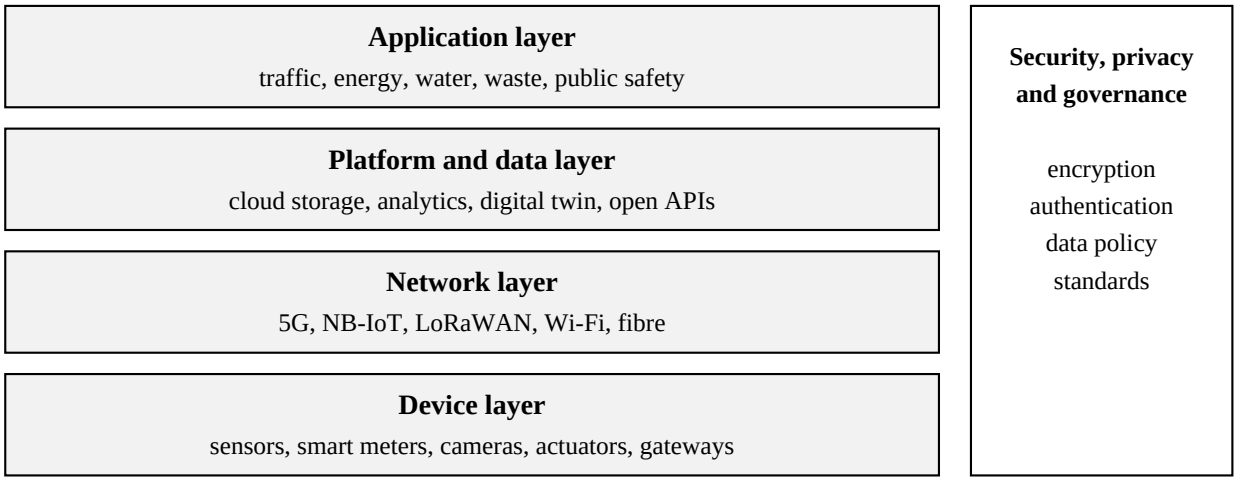


Figure 2. Layered architecture of a smart-city IoT system.

3. Main application areas

Table 1 lists the service areas where IoT is used most, what is installed and what cities hope to get out of it.

Area	Typical devices and data	Typical benefit
Traffic and mobility	Road and camera sensors, connected signals, parking sensors, transit GPS	Less congestion, less time spent looking for parking, live transit information
Energy and lighting	Smart meters, adaptive LED street lights, grid sensors	Lower consumption, demand response, easier use of renewables
Water and environment	Flow and leak sensors, soil moisture and rain gauges, air and noise monitors	Less leakage and irrigation waste, pollution alerts
Waste	Fill-level sensors in bins, vehicle tracking	Better collection routes, lower fuel cost
Public safety	Cameras, fire and flood sensors, emergency vehicle priority	Faster response, early warning of disasters
Buildings and health	Building energy systems, remote patient monitoring	Energy savings, better care for older people

Table 1. Main smart-city IoT application areas.

4. Case studies

4.1 Barcelona, Spain

Barcelona started rolling out connected technology in 2012 across public transport, parking, street lighting and waste. Its smart city programme picked 12 areas to work on and ran 22 programmes made up of 83 projects [3]. Adaptive LED lighting with sensors saved about 30% of lighting energy. Remote control of irrigation, installed in 68% of public parks, improved water conservation by around 25% and saved roughly US$555,000 a year [3]. The city estimates that overall it saved about US$58 million on water, earns an extra US$50 million a year from parking, saves US$37 million a year on lighting and gained 47,000 jobs [3]. These are the city's own estimates and have not been independently audited.

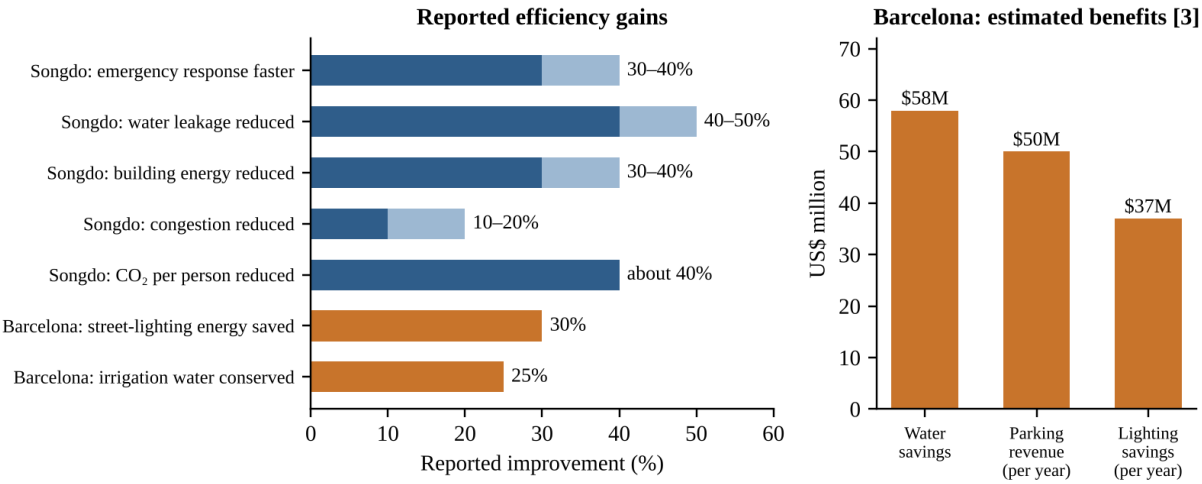


Figure 3. Reported results from Barcelona [3] and Songdo [7]. For ranges, the dark part of the bar marks the low end and the light part the high end. All figures are self-reported.

Not everything worked. A pilot of 100 commercial parking sensors in the 22@Barcelona district was never scaled up, and the city built its own mobile parking app, L'apparkB, instead [4]. One local academic described the risk as a technology push rather than a pull, meaning the technology should come from real needs [4].

4.2 Singapore

Singapore launched its Smart Nation initiative in 2014 and treats the country as a living laboratory for urban solutions, with sensors on things like bus fleets and public housing [5][6]. Its best-known project is Virtual Singapore, described as the first digital twin of a whole country. It is a detailed 3D model that can take in live sensor data and be used to simulate scenarios, plan transport, prepare for disasters and monitor the environment [5]. The Singapore Land Authority is now building a national subsurface twin as well [14]. The lesson here is the value of one shared data foundation led by government. That is easier to do in a small city-state than in a large country with many authorities.

4.3 Songdo, South Korea

Songdo was planned as a ubiquitous city from the start, on reclaimed land near Incheon Airport. A 2026 case study from the ITU's U4SSC initiative follows a phased programme from 2003 to 2030 that links IoT and AI across transport, buildings, energy, water, waste and public safety. It rests on city-wide fibre, LEED-certified buildings, district heating and cooling, autonomous buses and pneumatic waste collection. The reported results include 30-40% lower building energy use, about 40% lower CO_2 per person, 10-20% less congestion, more than 70% of waste recycled, 40-50% less water leakage and 30-40% faster emergency response [7].

Researchers have been more critical. Huh and colleagues argue that technology firms are badly placed to lead smart-city projects, because ICT is only a small part of the total budget and its products wear out much faster than buildings do [8]. Mullins found that Songdo's technology-first approach left out social needs and coped poorly with the messiness of real cities [9]. Put together, the two views suggest that integration works best when it is planned from the beginning, but that a city also needs people and community, not just sensors.

	Barcelona	**Singapore**	**Songdo**
Approach	Retrofit of an existing city, project by project	National data platform led by government	New city planned end to end
Main IoT use	Lighting, irrigation, parking, waste	Digital twin and sensor networks	Buildings, transport, waste and water
Strength	Measurable savings and a local tech industry	One shared 3D data base for planning	Consistent infrastructure from day one
Lesson	Start from needs, not products	Central data governance helps scaling	Technology alone does not make a liveable city

Table 2. Comparison of the three cities.

5. Technologies shaping the next decade

Edge AI. Processing data where it is produced lets devices make quick decisions on their own and reduces the amount of data sent to the cloud. Reviews of the field still list setup cost, device security and getting different devices to work together as open problems [11].

5G and 6G. 5G already offers high reliability, low delay and the ability to connect huge numbers of devices. Research on 6G aims for even higher speeds and near-zero delay, with AI built into network management, and smart-city infrastructure is one of its stated targets. The same research warns that 6G will widen the security and privacy attack surface [10].

Digital twins. Singapore shows how a live virtual copy of a city lets planners test traffic, flood or construction scenarios before touching the real thing [5]. I expect twins to grow from single buildings and districts to whole cities covering several utilities at once.

AI orchestration and blockchain. Recent research looks at AI, 6G, blockchain and the metaverse together as ways to make cities more resilient, while also raising questions about ethics and the digital divide [12]. In practice blockchain is mostly proposed for tamper-proof data sharing and device identity between organisations.

Sustainability targets. Cities increasingly justify IoT spending by energy, water and carbon goals, and they report results against them, as Barcelona and Songdo do [3][7].

6. Challenges and how to reduce them

The features that make a smart city useful are the same ones that create risk: sensors everywhere, shared data and automation. The Songdo study names cost, interoperability, cybersecurity, privacy, public trust and technology becoming obsolete as the main difficulties [7]. Table 3 pairs each with a practical response.

Challenge	Why it matters	What helps
Cybersecurity	Millions of cheap devices give attackers many entry points, and a hacked signal or meter affects public services [10]	Device identity and encryption, secure firmware updates, network separation, constant monitoring
Privacy	Cameras and location data can turn monitoring into surveillance [13]	Collect less data, anonymise at the edge, set retention limits, independent oversight
Interoperability	Vendor-specific systems create silos and lock-in	Open standards and APIs, modular design, buying only open-data products
Cost and obsolescence	Electronics age faster than city infrastructure [8]	Phased roll-out, upgradeable hardware, budgeting for the whole lifecycle
Equity and trust	Benefits may miss poorer areas, and tech-led plans can ignore social needs [9]	Involve residents, design for inclusion, publish results and open data

Table 3. Main challenges and ways to reduce them.

7. Outlook and recommendations

Looking across the three cities, a few points keep coming up. First, start with the problem and not the product. Barcelona's parking pilot shows that a sound sensor is wasted if it does not meet a real need [4]. Second, build a shared data foundation, since Singapore's twin is useful mainly because many agencies use the same platform [5]. Third, plan security and privacy from the first day, especially as 5G and 6G widen the attack surface [10]. Fourth, build in phases and keep parts replaceable, so that short-lived electronics can be swapped without rebuilding the infrastructure around them [7][8]. Finally, involve residents, because public trust and community life decide whether the technology is accepted and used [9].

A city should also judge its programme by outcomes, not by the number of devices installed. The results reported above give a ready scorecard: energy use, water use and leakage, congestion, emergency response time and recycling rate. All of these figures are self-reported by the cities and programmes involved, so independent measurement would make them more convincing.

For fast-growing cities in South Asia, Barcelona's retrofit route is the encouraging one, because it does not require starting from scratch. Small, low-power deployments in traffic, water, waste and street lighting can give early, measurable benefits and build the data base for later expansion. This last point is my own interpretation and is not a finding from the cited sources.

8. Conclusion

IoT is becoming the nervous system of the modern city, and forecasts suggest investment will keep growing strongly to 2030. Barcelona, Singapore and Songdo all show real gains in energy, water, mobility and planning, but they also show that technology on its own is not enough. Edge AI, 5G and 6G and digital twins will shape what comes next. Whether they succeed will depend on how securely, openly and fairly they are used for the people who live in the city.

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